Registers

A/PSW	0x 00 44
BC	0x 00 00
DE	0x 00 00
HL	0x 45 11
SP	0x FF FF
PC	0x 00 2F

Flags

S	☐
Z	☑
AC	☐
P	☑
C	☐

```
1  LXI H,4510H
2  MOV E,M
3  INX H
4  MOV D,M
5  INX H
6  MOV C,M
7  INX H
8  MOV B,M
9
10 MOV A,E
11 ADD C
12 STA 4518H
13 MOV A,D
14 ADC B
15 STA 4519H
16 MVI A,00H
17 ADC A
18 STA 451AH
19
20 LXI H,4510H
21 MOV A,M
22 SUB C
23 STA 451BH
24 INX H
25 MOV A,M
26 SBB B
27 STA 451CH
28 MVI A,00H
29 ADC A
```

Machine Code | LED Array | Metrics

Machine Code

Line	Address	Machine Code	Source Code
1	0x0	21 10 45	LXI H,4510H
2	0x3	5E	MOV E,M
3	0x4	23	INX H
4	0x5	56	MOV D,M
5	0x6	23	INX H
6	0x7	4E	MOV C,M
7	0x8	23	INX H
8	0x9	46	MOV B,M
9			
10	0xA	7B	MOV A,E
11	0xB	81	ADD C
12	0xC	32 18 45	STA 4518H
13	0xF	7A	MOV A,D
14	0x10	88	ADC B
15	0x11	32 19 45	STA 4519H
16	0x14	3E 00	MVI A,00H

Displaying Memory Locations from 0x4510 to 0x451D

Double click the value to edit then press Enter to save the value or Tab to edit the next location.

0x4510	34
0x4511	12
0x4512	78
0x4513	56
0x4514	00
0x4515	00
0x4516	00
0x4517	00
0x4518	AC
0x4519	68
0x451A	00
0x451B	BC
0x451C	BB
0x451D	01

```
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Given $X = 1234H$

$Y = 5678H$

(i) Addition :

$$\begin{array}{r} 1234H \\ +\ 5678H \\ \hline 68ACH \end{array}$$

Sum = 68 ACH

Carry = 00H

(ii) Subtraction

$$\begin{array}{r} 1234H \\ -\ 5678H \\ \hline BBBCH \end{array}$$ (2's complement)

Difference = BB BCH

Borrow = 01H